Energy Conversion and Management 270 (2022) 116202

Contents lists available at ScienceDirect

Energy Conversion and Management

journal homepage: www.elsevier.com/locate/enconman

Computational modeling of a PV-PCM passive cooling system during a day–night cycle at arid and semi-arid climate zones

Felipe A. Díaz [a,*], Nelson O. Moraga [a], Roberto C. Cabrales [b]

[a] Departamento de Ingeniería Mecánica, Universidad de La Serena, Benavente 980, La Serena, Chile
[b] Instituto de Investigación Multidisciplinario en Ciencia y Tecnología, Universidad de La Serena, Raúl Bitrán, La Serena 1305, Chile

ARTICLE INFO

Keywords:
Photovoltaic panel
Phase change material
PV cooling
Passive cooling
Enhanced conduction model

ABSTRACT

This work investigates the use of phase change materials (PCMs) as a passive cooling system to increase the electric power generation of photovoltaic panels (PV) by a reduction of the temperature of the panels. This study aims to evaluate different configurations of PV-PCM cooling systems, by changing the phase change material, to achieve a high efficiency in a silicon PV. Each configuration is studied for three day–night cycles under seasonal variable atmospheric conditions. Two Chilean cities with a semi-arid climate and arid climate – Vicuña and Calama, respectively – were selected as the study locations. The development of the PV-PCM improved system is carried out by solving the unsteady thermal energy equation with computational simulations using the finite volume method (FVM). An Enhanced Conduction Model (ECM) was used to include the convective heat transfer in the melted PCM. The ECM was validated against numerical values of a PV-PCM system finding a maximum deviation for the PV frontal temperature of 1.23 °C. The numerical model was validated with experimental and numerical results of a PV-PCM system during a 24-hour cycle with time-varying atmospheric conditions. The results obtained indicate that the use of phase change materials as a cooling system decreases up to 17.5 °C the temperature at the PV, diminishing the losses of efficiency and increasing the generation of electricity. The electrical production in one year increases by 5.8% for Vicuña and 4.5% for Calama when a layer of $CaCl_2$–$6H_2O$ with a thickness of 40 mm is used.

1. Introduction

In 2019, 26,636 TWh of electrical energy were consumed [1], with and increment of 1.2% over than the electrical consumption of 2018 [2]. The 64.2% came from fossil fuels (coal, oil, and natural gas) which are non-renewable sources of energy contributing to the global warming. The harmful effect of hydrocarbons on nature has motivated the use of photovoltaic panels as renewable alternatives to replace contaminant energy sources. However, the efficiency of the electric generation in the available technology decreases when the PV panel temperature increases, with a relevant negative impact on the energy production [3]. In order to solve this problem, new technologies based on the use of phase change materials (PCM) as a passive cooling system have been developed to reduce the operation temperature increasing the energy production of the solar panels [4].

Many successful attempts to improve the performance of PV-PCM system have been reported recently in the literature. Khanna et al. (2018a) performed optimization studies of PV-PCM with RT 25 HC by numerical simulations finding that PCM with melting point near to ambient temperature maintain lower PV temperatures, but required higher thickness [5]. Khanna et al. (2018b) also studied finned PV-PCM systems observing that increasing the cavity thickness extend the panel cooling period [6]. Kamkari et al. (2014) investigated the effects of the inclination angle of the panel finding that a closure with horizontal orientation has more than two times higher heat transfer rate (heat transfer between two surface per unit of time) than a closure with vertical orientation [7]. Nasef et al. (2022) experimentally studied parallel and staggered patterns of PCM for thermal management concluding that staggered arrays have higher average heat transfer coefficients than parallel arrays [8]. Nižetić et al. (2018) evaluated pork fat as a potential low cost substitute for the PCM finding that this system allowed to obtain similar results than the PV-PCM system [9]. Akshayveer et al. (2021) proposed a bifacial PV-PCM system that utilizes 87% more solar irradiation than a conventional PV-PCM system [10]. Emam et al. (2018) extended the PV-PCM system to a concentrator photovoltaic one using up to 15 cm of PCM in configurations with single and multiples PCMs [11]. In order to increase the PV-PCM system efficiency, researchers have used nanoparticles in the PCM to increase the thermal conductivity enhancing the efficiency of the PV-PCM system [12]. Nada

* Corresponding author.
E-mail addresses: felipe.diazm@userena.cl (F.A. Díaz), nmoraga@userena.cl (N.O. Moraga), rcabrales@userena.cl (R.C. Cabrales).

https://doi.org/10.1016/j.enconman.2022.116202
Received 7 June 2022; Received in revised form 1 September 2022; Accepted 2 September 2022
Available online 14 September 2022
0196-8904/© 2022 Elsevier Ltd. All rights reserved.

Nomenclature

c_p, $\bar{c_p}$	Specific and apparent heat capacity, J kg^{-1} °C^{-1}
d	Thickness of the liquid domain, m
f_{pc}	Liquid fraction
g	Gravitational acceleration, ms^{-2}
h	Convective coefficient, W m^{-2}
H/d	Height and thickness of the liquid domain ratio
I	Solar irradiation, W m^{-2}
k	Thermal conductivity, W m^{-1} K^{-1}
L_h	Latent heat, J kg^{-1}
L_{ch}	Characteristic length of the geometry, m
q''	Heat flux, W m^{-2}
S_{glass}	Source term of the glass heat generation, W m^{-3}
S_{PV}	Source term of the photovoltaic panel heat generation, W m^{-3}
T	Temperature, °C
T_m	Melting temperature, °C
T_{air}	Air temperature, °C
T_g	Ground temperature, K
T_{sky}	Clear sky temperature, K
w	Wind velocity, ms^{-1}
x	Coordinate axis, m

Greek letters

α	Absorptivity
β	Volumetric expansion coefficient, K^{-1}
ΔT	Liquid–Solid temperature range
η	PV efficiency, %
μ	Dynamic viscosity, Pa s
ν	Kinematic viscosity, m^2 s^{-1}
ρ	Density, kg m^{-3}
σ	Stefan–Boltzmann constant, 5.67×10^{-8} W m^{-2} K^{-4}
θ	Angle of inclination, °
θ	Critical angle of inclination, °
ε	Emissivity

Adimensional Numbers

Gr	Grashof Number
Nu	Nusselt Number
Pr	Prandtl Number
Ra	Rayleigh Number

Subscripts

b	Back surface
l	Liquid phase
PV	Photovoltaic panel
s	Solid phase
u	Upper surface

et al. (2018) studied a PV-PCM/Al_2O_3 nanoparticle system and found that the PV panel temperature decreased by 10.6 °C and its efficiency increased by 13.2% [13]. Abdelrahman et al. (2019) improved the PV-PCM systems adding nanoparticles of Al_2O_3 obtaining a temperature reduction up to 46.3% respect to the basic PV-PCM equipment [14]. Al-Waeli et al. (2017) studied by experiments the performance of nano-PCM in a PV panel originated by the convective heat transfer in the nanofluid circulation [15]. All these studies positioned the PV-PCM system as an effective alternative to increase the electric production.

The modeling of PV-PCM system is a multi-physics problem that requires solving the heat transfer, fluid dynamics, phase change, and the instantaneous position of a moving solid front. All these phenomena add complexity to the numerical studies of the PV-PCM with an increased computational time and design costs. The increment in cost occurs by the optimization process to select the best configuration for different climates that involves the variation of parameters as the type of PCM and its thickness. Due to this, one-dimensional models has been used to reduce the computational time allowing increasing the numbers of cases to study and reducing the computational cost. Brano et al. (2014) proposed a simplified PV-PCM one-dimensional model validated against experimental measurements [16]. In the same line, Arıcıet al. (2018) used this model to study several PCMs to find the optimal configuration of the PV-PCM system for a city in Italy [17]. Velmurugan et al. (2021) proposed a simplified heat transfer model via a thermal resistance circuit for a PV panel with multiple PCMs [18]. However, these simplified one-dimensional models neglected the convective heat transfer of the PCM at the liquid phase. This approach is valid when the conductive heat transfer predominates over the convective heat transfer at Grashof number, Gr < 10^3, otherwise reduces the accuracy of the results [19]. Elsheniti et al. (2020) proposed an Enhanced Conduction Model (ECM) based on a one-dimensional model that includes the convective heat transfer obtained from Nusselt number correlations [20]. This model reduces the computational time up to three orders of magnitude with a deviation of 0.3 °C respect to the CFD results that includes natural heat convection. Nevertheless, the use of the ECM model along the PV panel layers, with time-varying heat radiation and convection losses, and variable day–night atmospheric conditions to find the best PCM configuration has not been fully explored. The literature review reveals a gap in the study of PV-PCM systems that simultaneously include the analysis of the relevant parameters affecting the power generation for different climate zones. This paper incorporates the hourly variation of the atmospheric conditions for a typical day–night cycle for summer, autumn, winter, and spring; a comparison of the performance of the PV-PCM system for a semi-arid and arid climate zones; and the efficient performance of a reduced one-dimensional unsteady heat transfer model with phase change and heat convective and radiation boundary conditions.

The goal of this paper is the study on boosting the power output of PV systems by passive cooling by using PCM's with hourly variation of the heat convection and radiation boundary conditions in a day–night 24 h cycle, that incorporates the time-variable ambient conditions at semi-arid and arid climate zones. The analysis was applied to Vicuña and Calama, Chile, that are located in the north of Chile, one of the regions with the highest solar radiation available worldwide. The objective is to find the best configuration of a PV-PCM system, out of seven alternatives of suitable PCM's, under time-varying atmospheric conditions for cities located at semi-arid and arid regions by computational modeling. The numerical simulations are performed using a one-dimensional finite volume approach with the ECM and a simplified PV physical model that incorporates time-varying boundary conditions of convection and radiation. The simulations are solved using an in-house code developed in Julia. The results included the presentation of estimated yearly results for semi-arid and arid climate zones of: the reduction of PV temperature and the increase of electrical energy production for a PV-PCM system; and the analysis of time-varying atmospheric conditions (day–night cycle and season variation) on the performance of seven PCMs, each one with six thicknesses.

This articles distinguish from others by the following features: (a) the use of computational modeling by the finite volume method based on an enhanced unsteady heat conduction model that includes time-varying boundary condition by radiation and convection; (b) the incorporation of the variation in time of irradiation and velocity and temperature of wind, during 24 h of a day–night cycle, for the four seasons at a semi-arid and arid climate zones; (c) the assessment of the thickness for seven existing PCMs on the performance of PV-PCM systems.

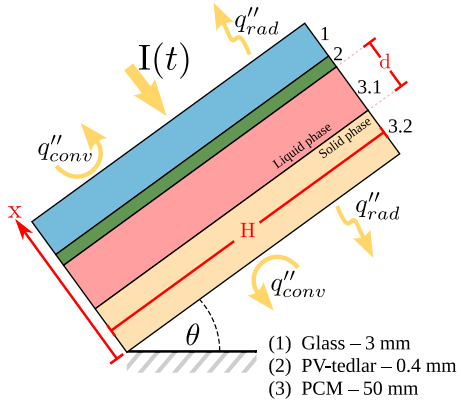


Fig. 1. Photovoltaic panel with Phase Change Material passive cooling system (PV-PCM).

2. Physical model

A single crystalline Photovoltaic panel (PV) is considered for the PV-PCM system. However, due to the manufacture procedure of PV, with multiple thin layers which added complexity to the physical model, instead of a full model, we used the simplified model proposed by Arıcıet al. (2018) [17]. This model includes four layers: glass (3 mm), silicon cell, Tedlar®-PVF (0.4 mm), and the PCM (50 mm) (Fig. 1). Due to the extremely thin thickness of the silicon layer in comparison with the others materials, the silicon layer is included in the tedlar domain as a PV-tedlar layer. Therefore, the incident solar energy that the panel does not transform in electricity is absorbed as heat being accounted for by a volumetric heat source term (S_{PV}) in the control volume of the PV-tedlar layer in contact with the glass of the panel.

In this research were performed studies for PV-PCM system configurations with one PCM for two cities with semi-arid and arid climates. The study considered seven PCMs for the PV-PCM configurations with one PCM. The selected PCMs were: RT-21, RT-25, RT-35, n-Octadecane (nOct), Calcium Chloride Hexahydrate ($CaCl_2$–$6H_2O$), Eutectic of capricepalmitic acid (CA), and Eutectic mixture of capric–lauric acid (CL). The PCM thickness increased from 5 mm up to 50 mm and the height used for the aspect ratio (H/d) was 50 mm for all cases. Three parameters were considered for the selection of the PCM: (1) The one being used in previous works for cooling PV systems; (2) A range of melting temperature from 18 °C to 35 °C; and (3) High value of the latent heat. Table 1 indicates the thermal properties of the PV materials and PCMs used in the numerical simulations. The apparent specific heat model required a temperature difference between the liquid temperature (T_l) and the solid temperature (T_s) in which the liquid to solid phase transformation occurs. In this work ΔT were equal to 1.0 °C and hence the values used in the simulations were $T_l = T_m + 0.5$ °C and $T_s = T_m - 0.5$ °C.

3. Mathematical models

The mathematical model uses the ECM to include the convective heat transfer of the melted PCM in the one-dimensional energy equation. This one-dimensional mathematical model solves the PV-PCM system has a similar accuracy to a two-dimensional CFD model, with a reduction of three orders of magnitude in the time of computation [20]. In order to simplify the description of the mathematical model for the PV-PCM system, this section is split in two subsections: (1) the conduction model used for the PV module and (2) the enchanted conduction model — ECM used to predict the heat transfer in the PCM.

3.1. Conduction model: PV module

The mathematical model for the PV module involves the one-dimensional transient diffusion energy transport equation (Eq. (1)). The thermodynamic properties of the materials used in the PV module are considered constant with temperature

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + S. \tag{1}$$

where the heat generation by unit volume in the PV-tedlar layer is calculated from the time-varying solar irradiation I(t) by,

$$S_{glass} = \alpha_u I. \tag{2}$$

$$S_{PV} = (1-\rho_{PV})I - \eta_{PV} I - S_{glass}. \tag{3}$$

where the absorptivity of the glass upper surface (α_u) [21] and the PV top surface reflectivity (ρ_{PV}) are equal to 0.05 [5].

The PV efficiency (η_{PV}) is computed from the Eq. (4) [27]. The nominal efficiency of single crystalline panels is generally smaller that 22% [28], and hence that a value of 20% was chosen for the calculations. The panel studied has a temperature coefficient of 0.005 $°C^{-1}$ and a solar irradiance coefficient of 0.85.

$$\eta_{PV} = 0.20(1 - 0.005(T_{PV} - 25) + 0.085\ln(I/1000)) \tag{4}$$

The conjugate boundary conditions are employed at each interface of different layers, based on continuity in the value of temperature and on the energy balance:

$$T_i(x,t) = T_j(x,t); \quad -k_i\left(\frac{\partial T_i}{\partial x}\right) = -k_j\left(\frac{\partial T_j}{\partial x}\right) \tag{5}$$

3.2. Phase change model: Enhanced conduction model (ECM)

The PCM mathematical model involves the 1D transient diffusion energy transport equation with the apparent heat capacity phase change model (Eq. (6). The phase change model is implemented with a liquid phase change fraction that changes linearly with temperature [29] (Eq. (7)). The temperature dependent density and heat capacity are presented in Eqs. (8) and (9).

$$\rho\bar{c_p}\frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) \tag{6}$$

$$f_{pc}(T) = \begin{cases} 0 & \rightarrow \quad T < T_s \\ \frac{T-T_s}{T_l-T_s} & \rightarrow \quad T_s < T < T_l \\ 1 & \rightarrow \quad T > T_l \end{cases} \tag{7}$$

$$\rho(T) = \begin{cases} \rho_s & \rightarrow \quad T < T_s \\ \rho_s + f_{pc}(T)(\rho_l - \rho_s) & \rightarrow \quad T_s < T < T_l \\ \rho_l & \rightarrow \quad T > T_l \end{cases} \tag{8}$$

$$c_p(T) = \begin{cases} c_{ps} & \rightarrow \quad T < T_s \\ c_{ps} + f_{pc}(T)(c_{pl} - c_{ps}) + L\frac{df_{pc}(T)}{dT} & \rightarrow \quad T_s < T < T_l \\ c_{pl} & \rightarrow \quad T > T_l \end{cases} \tag{9}$$

The 1D energy equation (Eq. (6)) incorporates the transient, diffusive, and source terms. To include the convective heat transfer that occurs in the PCM when is in liquid phase, the Enhanced Conduction Model (ECM) was implemented. This model multiply the PCM thermal conduction of the liquid phase (k_l) by a Nusselt Number, Nu, to include the convective heat transfer (Eq. (10)):

$$k(T) = \begin{cases} k_s & \rightarrow \quad T < T_s \\ k_s + f_{pc}(T)(k_l - k_s) & \rightarrow \quad T_s < T < T_l \\ k_l Nu & \rightarrow \quad T > T_l \end{cases} \tag{10}$$

The Nu is computed in terms of the angle of inclination (θ), PCM aspect ratio (H/d), and the dimensionless numbers Ra, Gr, and Pr (Eq. (11)), where Lch is the characteristic length of the liquid–solid

Table 1
Thermophysical properties of the PV layer materials and PCM used in the PV-PCM system.

Property	Glass	Tedlar	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
ρ, (kg m^{-3})									
solid	3000	1200	880	785	880	865	1710	870	890
liquid	–	–	770	749	760	770	1560	790	770
c_p, (kJ kg^{-1} K^{-1})									
solid	0.5	1.25	2.0	1.8	2.046	1.934	1.4	2.0	1.97
liquid	–	–	2.0	2.4	2.4	2.196	2.1	2.3	2.24
k, (W m^{-1} K^{-1})									
solid	1.8	0.2	0.2	0.19	0.2	0.358	1.08	0.14	0.143
liquid	–	–	0.2	0.18	0.2	0.148	0.56	0.14	0.139
L_h, (kJ kg^{-1} K^{-1})	–	–	155	232	157	243.5	191	173	168
μ, (Pa s)	–	–	0.00361	0.00179	0.00251	0.008	0.00287	0.00182	0.00433
β, (K^{-1})	–	–	0.001	0.001	0.001	0.00091	0.0005	0.00078	0.00067
T_m, (°C)	–	–	21	26.6	35	28	29.8	22.5	18.5
References	[21]	[21]	[22]	[22,23]	[24]	[11,25]	[11,26]	[11,26]	[26]

phase change in the cavity with PCM, corresponding to the length of the domain in liquid phase, that is equal to the variable length d in Fig. 1.

$$Pr = \frac{c_p \mu}{k}; \quad Gr = \frac{g\beta(T_{hot} - T_{cold})L_{ch}^3}{\nu^2}; \quad Ra = GrPr \tag{11}$$

The first step to compute the Nu by the ECM depends on the PCM angle θ, as shown in Eq. (12):

$$Nu(\theta) = \begin{cases} Nu_{Horizontal} & \rightarrow \quad \theta = 0° \\ Nu_{Inclined} & \rightarrow \quad 0° < \theta < 90° \\ Nu_{Vertical} & \rightarrow \quad \theta = 90° \end{cases} \tag{12}$$

The horizontal Nu correlation is based on the Ra [30]:

$$Nu_{Horizontal} = 1 + 1.44\left[1 - \frac{1708}{Ra}\right]^+ + \left[\frac{Ra^{1/3}}{18} - 1\right]^+ \tag{13}$$

where the brackets notation $[\,]^+$ mean that if the result is negative, it should be taken as zero. On the other hand, the vertical Nu correlation is found from the Ra, Pr and the PCM H/d ratio. According to the H/d of the PCM four correlations could be used to compute the Nu:

$$Nu_{Vertical} = \begin{cases} 0.18\left(\frac{Pr}{0.2+Pr}Ra\right)^{0.29} & \rightarrow \quad 1 < \frac{H}{d} < 2 \quad [31] \\ 0.22\left(\frac{Pr}{0.2+Pr}Ra\right)^{0.28}\left(\frac{H}{d}\right)^{-\frac{1}{4}} & \rightarrow \quad 2 < \frac{H}{d} < 10 \quad [31] \\ 0.42Ra^{1/4}Pr^{0.012}\left(\frac{H}{d}\right)^{-0.3} & \rightarrow \quad 10 < \frac{H}{d} < 40 \quad [32] \\ 1 & \rightarrow \quad \frac{H}{D} > 40 \quad [20] \end{cases} \tag{14}$$

The Nu for θ values between 0° to 90° is decomposed in three equations. When $H/d \geq 12$; $0° < \theta < 90°$; and $Ra < 1\times10^4$ the equation proposed by Hollands et al. (1976) is used [30]:

$$Nu_{Inclined} = 1 + 1.44\left[1 - \frac{1708}{Ra}\right]^+\left(1 - \frac{1708(\sin(1.8\theta))^{1.6}}{Ra\cos(\theta)}\right) + \left[\frac{(Ra\cos(\theta))^{1/3}}{18} - 1\right]^+ \tag{15}$$

In the case that $H/d < 12$, Ayyaswamy and Catton [33] proposed two equations based on a critical angle of inclination (θ_{cr}), this parameter depends on the H/d:

$$Nu_{Inclined} = \begin{cases} Nu_{Horizontal}\left(\frac{Nu_{Vertical}}{Nu_{Horizontal}}(\sin(\theta_{cr}))^{0.25}\right)^{\theta/\theta_{cr}} \rightarrow 0° < \theta < \theta_{cr} \\ Nu_{Vertical}\sin^{0.25}(\theta) \quad \rightarrow \theta_{cr} < \theta < 90° \end{cases} \tag{16}$$

The complete explanation of the ECM can be found on Elsheniti et al. (2020) work [20].

3.3. Initial and boundary conditions

The air temperature at the beginning of the problem is considered as the initial condition for all the materials of PV-PCM systems. However, to ensure that the solution is independent of the unknown uniform temperature initial condition each case of the study is simulated until the 72 h and only the results in the last 24 h are considered in this study.

$$T(x, 0s) = T_{air}(0s) \tag{17}$$

The boundary conditions at the top and bottom surfaces of the PV-PCM system included convective and radiative heat losses. h_u and h_b are the upper and lower convective coefficients, which have a value of h_{air} and half the value of h_{air}, respectively [34]. The emissivity factors used in the simulations were $\varepsilon_u = 0.95$ and $\varepsilon_l = 0.02$, respectively [21].

$$-k\frac{\partial T}{\partial x} = h_u(T_u - T_{air}) + \varepsilon_u\sigma(T_u^4 - T_{sky}^4) \tag{18}$$

$$-k\frac{\partial T}{\partial x} = h_b(T_b - T_{air}) + \varepsilon_l\sigma(T_b^4 - T_g^4) \tag{19}$$

where the shape factors for both ground and sky radiation were estimated to be equal to one, and the assumption that the backside of the system mostly sees the ground, at $T_g = T_{air}$ in K [17], due to the low values of the tilt angle ($\theta = 30°$ and 22°), for the two situations analyzed.

The heat convection coefficient (h) was computed by the equation proposed by Loveday and Taki [35], where w is the wind speed in ms^{-1}:

$$h_{air} = 8.91 + 2w \tag{20}$$

The sky temperature is estimated in terms of the air temperature as, [36]:

$$T_{sky} = 0.0552T_{air}^{1.5} \tag{21}$$

4. Atmospheric conditions

Two cities with semi-arid (Vicuña) and arid (Calama) climates in Chile were chosen as the places of study. The variation with time of the atmospheric conditions required as input parameters of the mathematical model – inclined solar irradiation, wind speed, and air temperature – were obtained from free online platforms. The atmospheric data for the semi-arid and arid city were provided by the Solar Explorer, developed by the Geophysics Department - Faculty of Physical Sciences and Mathematics - Universidad de Chile [37]. This tool used atmospheric models over the data of several atmospheric stations from the year 2004 up to 2016 to generate an output data with the atmospheric conditions. We select the semi-arid city, Vicuña-Chile, located at –30° 2′ 18", –70° 41′ 47" and the arid city, Calama-Chile, at

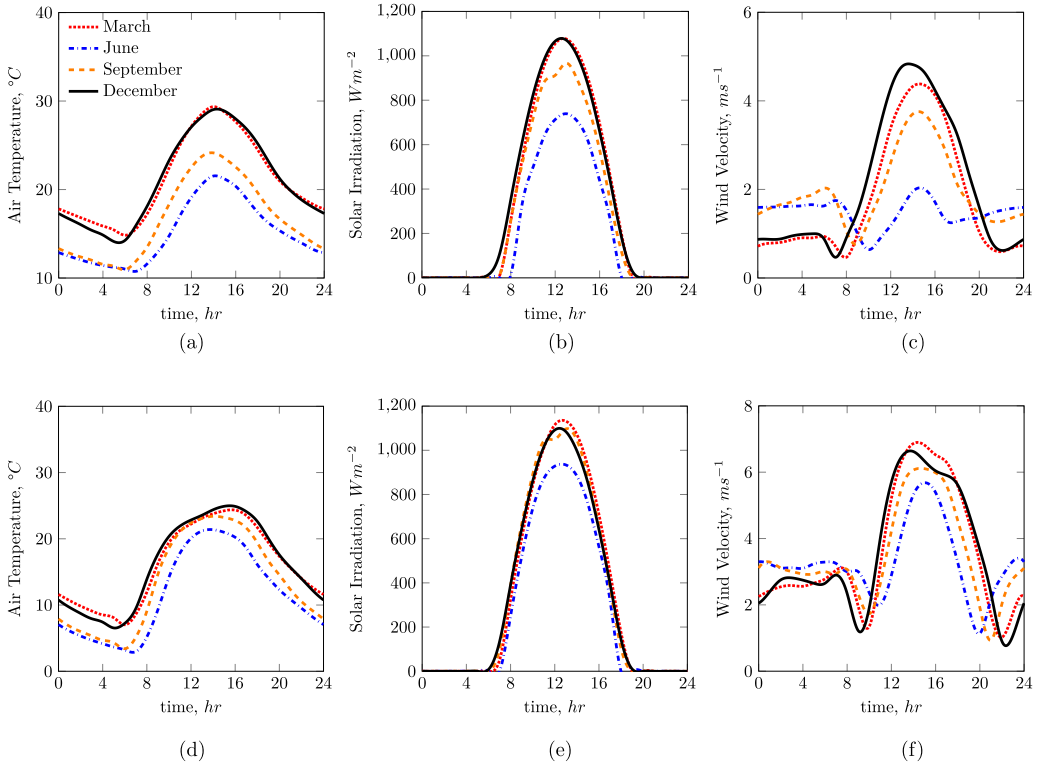


Fig. 2. Atmospheric conditions of the day–night cycle atmospheric conditions for the inclined solar irradiation, wind speed, and air temperature for semi-arid (a, b, and c) and arid (d, e, and f) cities at December, March, June, and September.

−22° 25′ 50", −68° 55′ 17". The angle of inclination of the PV coincides with the latitude of each city, 30° and 22° for the semi-arid and arid cities, respectively.

In order to study the behavior of the PV-PCM system at the most favorable and unfavorable scenarios in one year, this research used the atmospheric conditions from one month for each season: June for winter season, December for summer, and March and September for fall and spring, respectively. A representative atmospheric condition for the day–night cycle was obtained for each month in each city. The curves were computed as a mean day for each atmospheric variable in each month. The methodology used was by averaging the reported hourly value for every day of each month achieving as the result a representative day–night cycle for each month (Fig. 2).

The differences between the atmospheric conditions of the cities located at semi-arid and arid zones have an important impact in the selection of the PCM configuration for the PV-PCM system. Air temperature of the city at the semi-arid zone city varied from 10.8 °C to 29.0 °C, while in the city at the arid zone from 3.0 °C to 24.9 °C. The maximum air temperatures at the semi-arid city was higher than in the arid city. However, the thermal amplitudes of the arid city were between 2.8 °C to 7.7 °C higher than in the semi-arid city. The peak value of the inclined solar irradiation of the arid city was at the autumn season (March) with a value of 1132 $\mathrm{W\,m^{-2}}$, and with a minimum value of 934 $\mathrm{W\,m^{-2}}$ at the winter season (June). In the case of the semi-arid city, the maximum and minimum inclined solar irradiation were 1076 $\mathrm{W\,m^{-2}}$ in autumn and 739 $\mathrm{W\,m^{-2}}$ in winter, respectively. The city with arid climate had 26% more solar irradiation than the one in the semi-arid region, being this maximum difference at winter. The maximum values of irradiation in both cities were found in autumn. Lastly, the wind speed mean values of the semi-arid city were from 1.48 $\mathrm{ms^{-1}}$ to 2.08 $\mathrm{ms^{-1}}$, while in the arid city the values varied from 3.28 $\mathrm{ms^{-1}}$ to 3.54 $\mathrm{ms^{-1}}$. This information indicates that the mean wind velocity values of the arid city were approximately two times higher than the values at the semi-arid zone. We hypothesize that the large values of wind velocity of the arid city, will decrease the maximum PV temperature due to the expected larger convective heat losses on the PV.

5. Numerical procedure

The one-dimensional mathematical model was solved by the Finite Volume Method (FVM) by an in-house code programmed in Julia language [38]. The transient term was calculated with a second order Backward Difference (BDF2) and the diffusive term was evaluated with a central difference scheme. The spatial discretization was accomplished with a non-uniform structured grid and the time integration of all terms was calculated with an implicit scheme. The nonlinear boundary condition due to radiative heat transfer from top and bottom surfaces is linearized by the source-term linearization technique proposed by Patankar [39], in which the coefficients of the convective and radiative heat fluxes at the boundary are iteratively recalculated. The linear equation system was solved by a LU solver from the UMF-PACK library [40]. The temperature convergence criterion for each time step was imposing that the difference between the current and the previous iteration should be lower than 10^{-8}.

A mesh and time step size independence study was performed over the PV-PCM configuration of RT-25 with 50 mm thickness for the summer atmospheric conditions of the semi-arid city. A uniform time step of 1.0 s and a non-uniform mesh of 30 volumes for the glass layer, 12 volumes for the PV-tedlar layer, and 20 volumes per PCM mm were selected. The deviations of the maximum PV temperature and electric energy generation of the selected parameters against a mesh with 2420 volumes and a time step of 0.01 s were lower than 0.1%.

6. Model validation

The mathematical and physical models used in the numerical simulations included four major complexities: solid–liquid phase change,

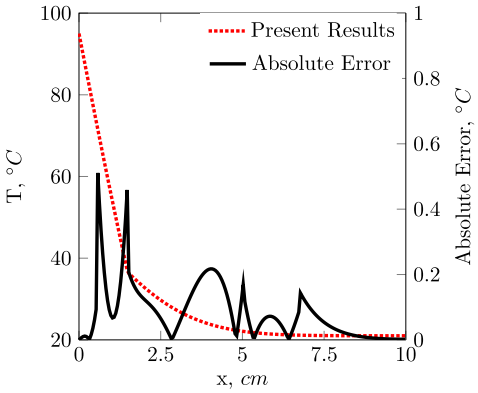


Fig. 3. Validation of the apparent heat capacity phase change model with the Stephan problem [41].

a PV model with multiple materials, time-varying radiation and convection boundary conditions during a day–night cycle, and the parameterization of the convective heat transfer (ECM) of the melted PCM. Therefore, to ensure the accuracy of the numerical model the validation procedure was carried out in a three steps procedure with incremental complexity.

6.1. Validation of apparent specific heat phase change model

The first problem validated the apparent heat capacity phase change model by comparing present results with those calculated by the analytical solution of the 1D Stephan problem reported by Solomon (1979) [41]. A semi-infinite wall of N-Eicosene paraffin wax is melted with a Stephan number equal to 0.48. The wax is initially at 21 °C, and the boundary conditions are: a constant temperature of 95 °C imposed at the left hot wall, and an adiabatic boundary condition at the right wall. The study is simulated for 3600 s, for a wax length of 10 cm. The comparison with the analytical solution indicates that the maximum deviation was 0.51°C for the temperature profile, and 0.1 mm for the instantaneous position of the solid–liquid interface 3. The solution of this problem allows the validation for the implementation of the heat capacity phase change model by the FVM to predict the solid–liquid phase change process.

6.2. Conduction phase change validation

The second study validates the simplified 1D PV-PCM model with the experimental and numerical results reported in the works of Brano et al. (2014) [16] and Arıcıet al. (2018) [17]. The physical situation consists in a PV-PCM system with 5 cm of PCM, and the time-varying atmospheric conditions at Palermo, Italy, during September 4, 2010. In this case, the natural heat convection on the melted PCM is neglected. Fig. 4 shows the comparison of the present results with the numerical and experimental results reported. It is found that the maximum PV temperature occurs at the 11 hrs with 3 min, achieving a value of 60.5 °C. The results exhibit the same tendency as those in the references. Furthermore, our results from 19 h up to 9 h, have a good agreement with the numerical results reported by Brano et al. (2014). The validation of the mean and maximum temperature of the PV panel are showed in Table 2. The maximum differences between the maximum PV temperature respect our results are: 1.4 °C respect to the experimental results of Brano et al. (2014), 2.2 °C from the numerical results of Brano et al. (2014), and 0.46 °C respect to the numerical prediction of Arıcıet al. (2018).

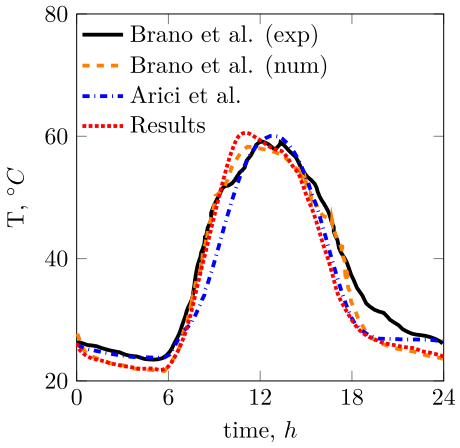


Fig. 4. Validation of PV-PCM simplify model with the results reported by Arıcıet al. (2018) [17] and Brano et al. (2014) [16].

Table 2
Validation of PV mean and maximum temperature of PV-PCM reduced model.

	T_{mean}, °C	T_{max}, °C
Results	35.4	60.5
Brano et al. (exp) [16]	37.6	59.1
Brano et al. (num) [16]	35.8	58.3
Arıcıet al. [17]	35.5	60.1

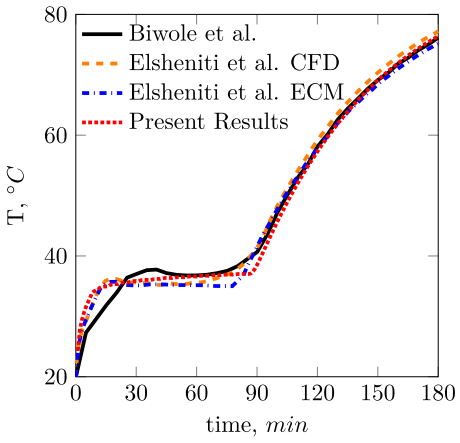


Fig. 5. Validation of ECM with the works of Biwole et al. (2014) [23] and Elsheniti et al. (2020) [20].

6.3. Enhanced conduction model (ECM) validation

The last study validates the ECM describing heat transfer and melting of a confined PCM inside two vertical plates of aluminum. This problem is reported by Biwole et al. (2013) [23] and used as validation in the works of Elsheniti et al. (2020) [20]. Fig. 5 presents the results of the evolution of temperature that shows a good agreement with the CFD and 1D ECM results reported in the work of Elsheniti et al. (2020). The temperature deviation at 180 min between our results is: 0.24 °C respect Biwole et al. (2013) results, and 0.86 °C and 1.23 °C in comparison with the CFD and 1D ECM results of Elsheniti et al. (2020), respectively. This study validates the FVM implementation of the ECM for the numerical study of PV-PCM systems.

7. Results and discussion

The PV-PCM system was evaluated using hourly reported data, for day–night cycles, of two cities in Chile, one located in an arid zone and

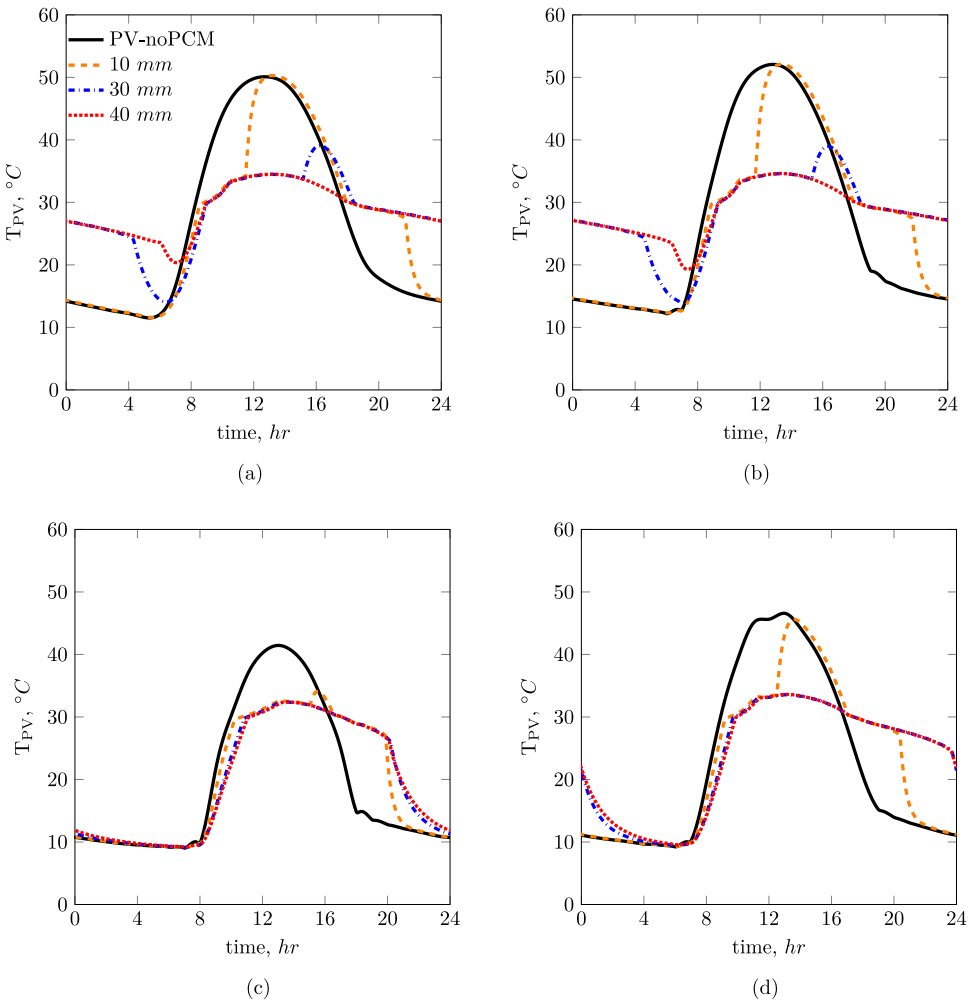


Fig. 6. Effect of thickness on maximum PV temperature in the PV-PCM system with $CaCl_2$–$6H_2O$ at semi-arid for (a) summer, (b) autumn, (c) winter, and (d) spring season.

the other in a semi-arid region. Four day–night cycles, one per each season, was used to select the optimal configuration for each location.

7.1. PV-PCM system

The PV-PCM system acts as a thermal sponge absorbing the large amounts of heat that the PV cell did not transform into electricity, without raising the PV temperature because the PCM requires large energy amounts to change from the solid to the liquid phase, with a small temperature interval. The PV-PCM system decreases the maximum PV temperature and provokes its displacement in time. Therefore, when the peak of solar irradiation reaches the panel, which is at a low temperature, the PV-PCM system achieves a high electric efficiency increasing the generation of electricity. The design of this system has two important parameters, which are critical in the thermal performance: the adequate thickness and the appropriate choice of the PCM. The amount of PCM has a direct impact on the PV performance, if the amount of PCM is too low the PV-PCM system could have a negative impact on the PV temperature and in the electricity production.

Fig. 6 shows the effect of the thickness of $CaCl_2$–$6H_2O$ in the PV temperature for the city in the semi-arid zone at the four seasons. The results for $CaCl_2$–$6H_2O$ at the semi-arid region show that the optimal thickness configuration is of 40 mm for summer and autumn, and 30 mm for winter and spring. PV-PCM systems of $CaCl_2$–$6H_2O$ with higher thickness are oversized and provide a negligible increment in the thermal performance. The reduction of the maximum PV temperature of $CaCl_2$–$6H_2O$ with a thickness between 20 and 30 mm, 30 to 40 mm, and 40 to 50 mm for summer season was 7.9, 4.6, and 0.0 °C, respectively. The rate of the cooling effect of the PV-PCM system respect to the PCM thickness is higher for the lower thickness configurations.

In the case of the arid climate city, Fig. 7 presents the maximum PV temperature of the PV-PCM system with $CaCl_2$–$6H_2O$ for the four seasons. The optimal thickness for the maximum decrease in the maximum PV temperature was 30 mm for the four seasons. The 50 mm thickness configuration increases the maximum PV temperature in 0.1 °C more than the 30 mm configuration for the summer and spring seasons.

The heat fluxes were analyzed to understand the role of the heat transfer mechanisms of the PV-PCM system. Fig. 8 shows the heat losses by convection and radiation and the gains by solar irradiation of the PV cells without PCM for the cities with semi-arid and arid climates at the warmest month. The heat generation due to the solar irradiation is the main source of heat in the PV-PCM system. This system has four main heat loss mechanisms contributing to cool down the PV panel: upper convective losses (qu''_{conv}), upper radiative losses (qu''_{rad}), lower convective losses (qb''_{conv}), and lower radiative losses (qb''_{conv}). However, in all scenarios the magnitude of the lower radiated losses is negligible. In the studies without PCM, the upper convective heat losses had the major impact on the cooling process of the PV panel, followed by the upper losses by thermal radiation and the lower heat convective losses, respectively. In the PV-PCM system with 50 mm of $CaCl_2$–$6H_2O$, the heat transfer by radiation has a major contribution than the convective heat losses for the city in the arid region. However, for this case, heat transfer losses by convection and radiation have a similar magnitude. Due to the liquid–solid phase change the PV-PCM system reduces the

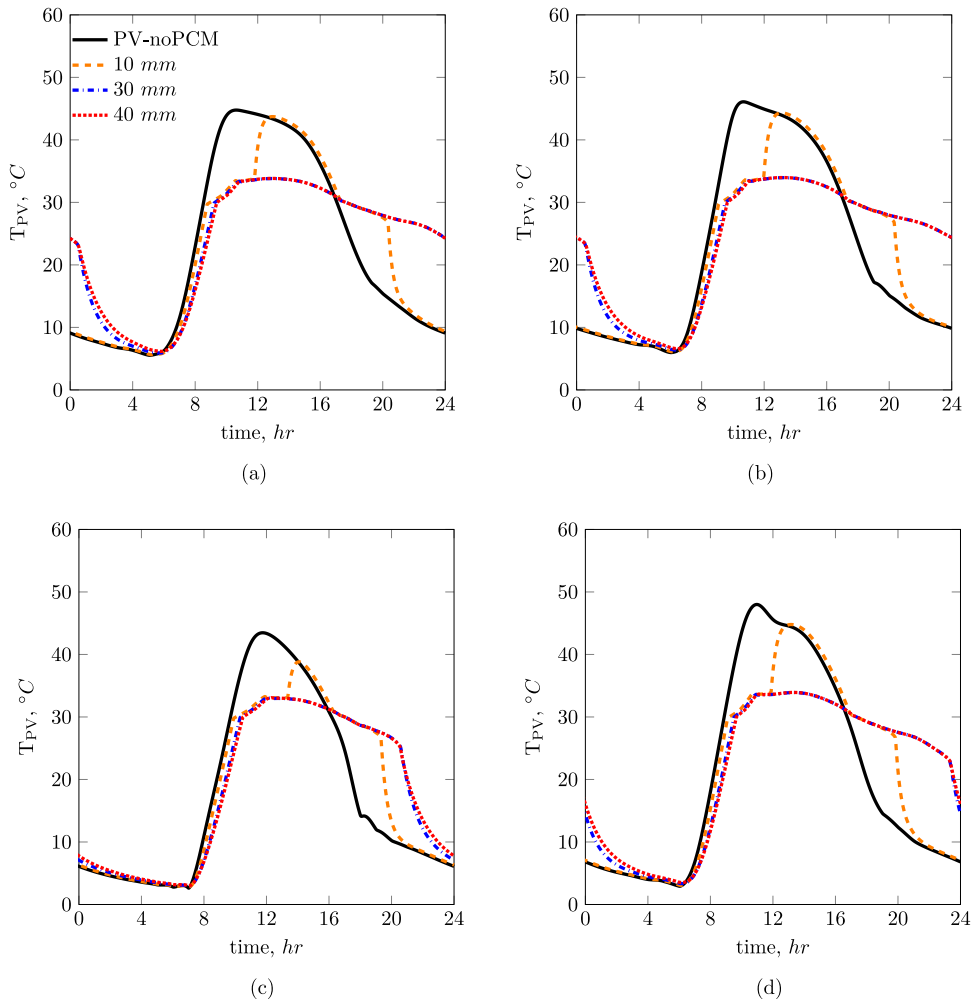


Fig. 7. Effect of thickness on maximum PV temperature in the PV-PCM system with $CaCl_2$–$6H_2O$ at arid for (a) summer, (b) autumn, (c) winter, and (d) spring season.

maximum PV temperature, then heat losses by convection and radiation are decreased. The main cooling mechanism on the PV-PCM system is the absorption of heat by the melting process of the larger thermal mass of the PCM during the phase change.

In summer, during a day–night cycle of 24 h in an arid climate, the performance by including time-varying losses in comparison with the calculations without time variations amounts (using mean values of air temperature and wind velocity) to 2 °C in the maximum temperature in the panel and 1.8% of over predicted energy generation for the PV system without a PCM, that reduces to 0.83 °C in the PV maximum temperature and a negligible variation in the energy generated when 40 mm of $CaCl_2$–$6H_2O$ are used in a PV-PCM system.

The PV-PCM system increases the electricity production by reducing the PV temperature during the day–night cycle in summer (Fig. 9). The maximum power output coincides with the maximum value of irradiation, the maximum PV temperature, and due to Eq. (4) also the maximum losses in electricity production. The PV-PCM system reduces the maximum temperature and displaces this peak value later on the day, achieving a lower temperature at the time when maximum value of solar irradiation occurs and hence generates more electricity.

7.2. Selection of PCM

The melting temperature and latent heat values of the PCM are the crucial parameters to select the optimal PCM for the PV-PCM system. However, the selection of the correct PCM and thickness will depend on the atmospheric conditions on each climatic zone. In zones with larger amount of solar irradiation and low values of wind speed, the PV temperatures will be higher, and therefore the electric production will be reduced. In these scenarios, larger thickness of a PCM with a high melting temperature will be required. On the other hand, in zones with low values of solar irradiation and high wind speeds, the values the PV temperature will be lower. However, in scenarios with low PV temperature and high heat convective losses, the PCM of the PV-PCM system must have a low melting temperature. In some cases, the use of a PV-PCM system could be oversized or inclusive unnecessary. Due to the selection of the appropriate PCM and its optimal thickness is a complex process, seven PCM with thickness between 5 and 50 mm were analyzed for the PV-PCM system for two cities with a semi-arid and an arid climates.

Maximum PV temperature and electricity production are the main parameters to compare the PV-PCM configurations investigated. Table 3 depicts the maximum temperature and electricity production for baseline scenarios at each climate conditions. Considering the electricity production, the system at the city in the arid region yearly generates 1.45 kWh/day/m^2, while the electricity generated by the system at the semi-arid region is 1.24 kWh/day/m^2. The yearly electricity production is calculated as the mean energy production of the four seasons studied. It is found that the seasonal variability in the maximum PV temperature between the extreme cases is 4.5 °C for the city at the arid region, and 10.7 °C for the one at the semi-arid zone. This seasonal variability for the city located at the semi-arid region is 2.4 times higher than in the city at the arid region.

Tables 4 and 5 shows the maximum PV temperature difference in a PV-PCM system respect to the base scenario for the semi-arid and arid

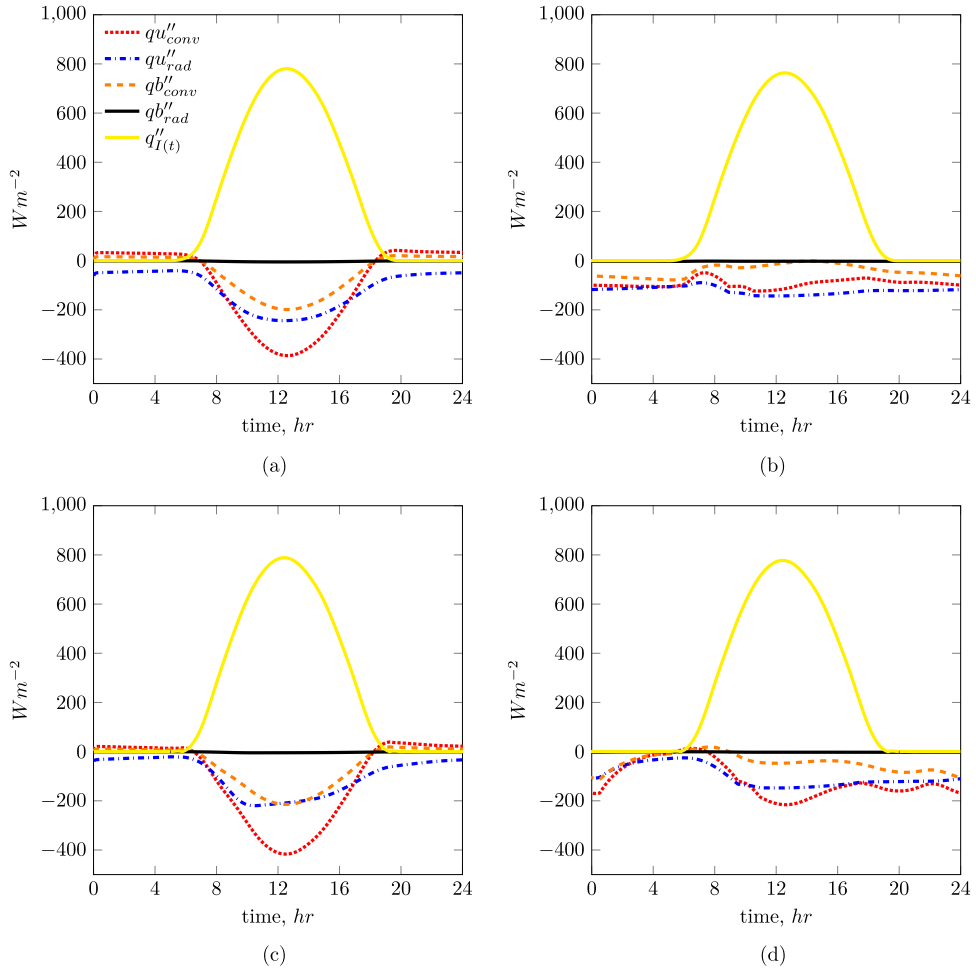


Fig. 8. Heat losses of the PV (left) and PV-PCM with 50 mm of $CaCl_2$–$6H_2O$ (right) at summer season at (a, b) semi-arid and (c, d) arid climate cities.

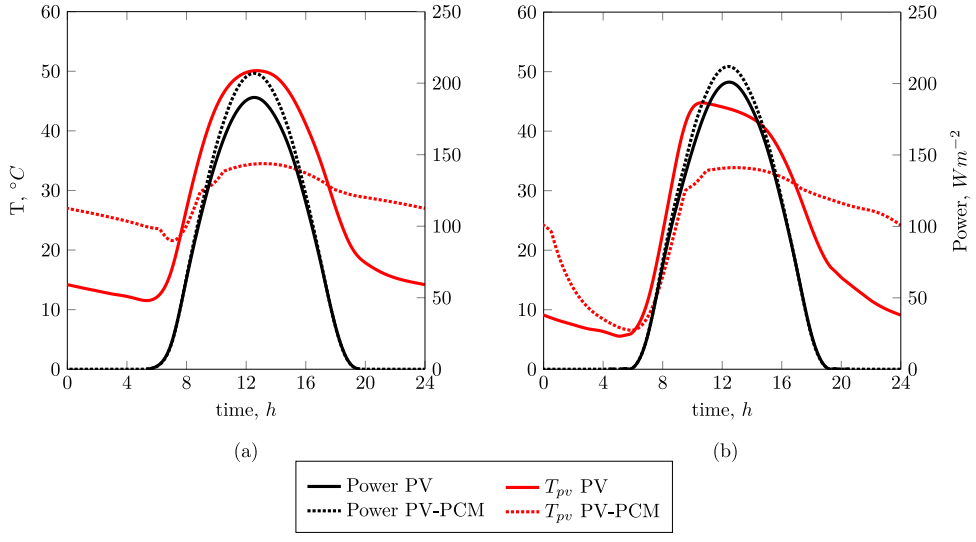


Fig. 9. Temperature and power of the PV system without and with 50 mm of $CaCl_2$–$6H_2O$ at the summer season at the (a) semi-arid and (b) arid climate city.

cities, respectively. The cases with layers of PCM which have lower thickness produced an increment of the maximum PV temperature. This is because the increment of the size of the PCM layer increases significantly the thermal resistance at the back of the PV panel. The thermal resistance of the PV panel is 3.67×10^{-3} m^2 °C W^{-1}. However when 10 mm of $CaCl_2$–$6H_2O$ were used the PV thermal resistance is increased by 9.26×10^{-3} m^2 °C W^{-1} and 1.79×10^{-2} m^2 °C W^{-1} for solidification and melting process. In these scenarios, the cooling effect of the PCM is insufficient to compensate the increment in the thermal resistance. Larger amounts of PCM are required to obtain a decrease in the maximum PV temperature. In the opposite case, a high thickness of the PCM results in an oversized system. After the use of a

Table 3
Base PV maximum temperature and daily electric generation for the semi-arid and arid cities at December, September, June, and March.

Month/Season	Semi-arid (Vicuña, Chile)		Arid - (Calama, Chile)	
	T_{max}, °C	E, kWhr/m^2/day	T_{max}, °C	E, kWhr/m^2/day
December/Summer	50.1	1.44	44.8	1.50
March/Autumn	52.1	1.39	46.1	1.53
June/Winter	41.4	0.89	43.5	1.25
September/Spring	46.6	1.24	48.0	1.51

Table 4
Semi-arid city results for maximum PV temperature reduction on PV-PCM system for seven PCM with thickness between 5 and 50 mm.

Seasons	mm	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
Summer	5	0.7	0.6	0.6	0.7	0.3	0.8	0.9
	10	0.8	0.7	0.7	0.7	0.2	0.9	1.1
	20	0.8	0.5	0.1	0.3	−3.1	0.8	1.1
	30	0.6	−1.1	−2.3	−1.9	−11.0	0.0	0.8
	40	0.2	−3.6	−6.9	−4.9	−15.6	−0.8	0.4
	50	−0.2	−5.0	−8.4	−6.4	−15.6	−1.7	−0.02
Autumn	5	0.7	0.6	0.6	0.6	0.3	0.7	0.9
	10	0.8	0.7	0.7	0.6	0.0	0.8	1.0
	20	0.7	0.2	−0.2	−0.02	−4.3	0.6	1.0
	30	0.3	−2.0	−3.2	−3.1	−13.1	−0.5	0.6
	40	−0.2	−5.1	−8.1	−6.5	−17.5	−1.7	0.0
	50	−0.8	−6.7	−10.1	−8.2	−17.5	−2.8	−0.6
Winter	5	0.4	0.4	−0.1	0.4	−0.5	0.5	0.6
	10	0.5	−0.1	−2.7	−0.7	−7.4	0.4	0.6
	20	−0.5	−6.5	−2.9	−9.1	−8.9	−2.0	−0.8
	30	−3.5	−9.7	−3.0	−9.2	−9.0	−8.0	−4.7
	40	−8.3	−9.7	−2.9	−9.1	−9.1	−12.7	−7.6
	50	−11.6	−9.7	−2.9	−9.1	−9.1	−12.6	−9.5
Spring	5	0.6	0.5	0.5	0.5	0.2	0.6	0.8
	10	0.7	0.6	0.5	0.5	−1.0	0.7	0.9
	20	0.5	−1.1	−3.7	−2.0	−11.2	0.1	0.6
	30	−0.7	−5.9	−6.2	−8.5	−13.0	−2.8	−1.2
	40	−3.0	−12.9	−6.2	−12.3	−13.0	−6.5	−2.7
	50	−4.3	−12.9	−6.2	−12.3	−13.0	−8.5	−4.0

certain amount of PCM, the cooling effect of the PV-PCM system stop increasing.

In the city at the semi-arid region maximum PV temperature results (Table 4), show that at summer, autumn, and spring, 40 mm, 40 mm, and 30 mm of $CaCl_2$–$6H_2O$ were the best thickness, respectively. However, at winter the optimal size was 40 mm of CA with a temperature reduction of 12.7 °C. The use of $CaCl_2$–$6H_2O$ in winter was inefficient, with a higher maximum temperature of 3.6 °C over than for the CA optimal configuration.

The results for the city at the arid region (Table 5) indicate that thickness of $CaCl_2$–$6H_2O$ between 30 and 50 mm had the best cooling performance for the PV-PCM system at the entire year. The optimal PCM thickness for summer, autumn, winter, and spring was 30 mm $CaCl_2$–$6H_2O$. However, for PCM with a thickness of 50 mm in winter, the use of CA reduced the maximum PV temperature 2.8 °C more than when $CaCl_2$–$6H_2O$ was chosen. In contrast to the semi-arid city results, the $CaCl_2$–$6H_2O$ can be used for the entire year with optimal results. Other good choices for the PV-PCM system for the city at the arid region are the RT-25 and the n-Octadecane.

Tables 6 and 7 shows the increment in electric generation for the PV-PCM systems respect to the baseline scenarios for the cities at semi-arid and arid regions. The maximum PV temperature in the PV-PCM system does not have a direct relationship with the electric generation in the majority of the cases. The systems with lower PCM thickness exhibited small increments of the maximum PV temperature, however, in the majority of these cases the electricity production was increased. This is explained because the PV-PCM system displaces the maximum PV temperature later on the day. Therefore, when the peak of solar irradiation occurs the PV panel has a lower temperature and hence a higher PV efficiency, generating a higher amount of electrical energy.

At the city in the semi-arid region, the increment of electricity generation of the PV-PCM system (Table 6) indicates that the optimal configurations for summer, autumn, and spring, were 40 mm, 40 mm, and 50 mm of $CaCl_2$–$6H_2O$. The selection for summer and autumn were the same as the selection by the maximum PV temperature. However, the selection of thickness for the spring season was 20 mm larger. At the season of winter, the optimal configuration was 50 mm of RT-21. This selection was with a different PCM than the previous one, were the optimal configuration – with a maximum PV temperature 1.0 °C lower – was 50 mm of CA. This finding indicates that lower maximum PV temperature could not be directly related to improvements in the electricity generation.

In the case of the city at the arid region, the selection of the optimal PV-PCM configuration by the increment of electricity generation (Table 7) exhibited similar results than at the city located in the semi-arid region. A PV-PCM system with 50 mm of $CaCl_2$–$6H_2O$ was the optimal configuration for summer, autumn, and spring season. This combination presented an increment of 20 mm of $CaCl_2$–$6H_2O$ respect to the optimal selection based on the maximum PV temperature. In the winter season, the PV-PCM system with 50 mm of CL was the optimal configuration, even though the 50 mm of RT-21 was only 0.1% less efficient. One difference with the results in winter of the city in semi-arid region was that the use of 50 mm of $CaCl_2$–$6H_2O$, instead of the optimal configuration, generated a lost in electricity generation of 1.0%, while at the semi-arid city this amount lost was 1.6%.

The electricity generation was used to select the seasonal optimal configuration. We follow the same approach to select the optimal PV-PCM system for the year. Table 8 presents the results of electricity generated in one year by PV-PCM systems with seven different PCMs at the cities located at semi-arid and arid regions. The yearly performance increase is calculated using the yearly electricity production. The yearly

Table 5
Arid city results for maximum PV temperature reduction on PV-PCM system for seven PCM with thickness between 5 and 50 mm.

Seasons	mm	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
	5	0.6	0.5	0.4	0.6	−0.1	0.7	0.8
	10	0.7	0.3	0.0	0.2	−1.0	0.6	0.9
	20	0.2	−0.8	−2.0	−1.2	−5.4	−0.1	0.3
Summer	30	−0.6	−2.8	−4.3	−3.9	−11.0	−1.5	−0.7
	40	−1.7	−6.5	−4.3	−10.1	−10.9	−3.4	−1.8
	50	−2.7	−10.6	−4.3	−10.1	−10.9	−5.1	−2.6
	5	0.6	0.4	0.2	0.4	−0.5	0.6	0.8
	10	0.5	−0.2	−0.6	−0.4	−1.8	0.3	0.6
	20	−0.4	−1.5	−2.6	−1.9	−6.5	−0.9	−0.3
Autumn	30	−1.4	−3.8	−5.4	−5.0	−12.1	−2.3	−1.5
	40	−2.6	−7.8	−5.4	−11.2	−12.1	−4.6	−2.7
	50	−3.8	−11.7	−5.4	−11.2	−12.1	−6.5	−3.6
	5	0.6	0.5	0.02	0.5	−0.7	0.6	0.8
	10	0.5	−0.4	−2.7	−0.8	−4.7	0.4	0.7
	20	−0.9	−4.6	−4.0	−6.1	−10.3	−2.3	−1.2
Winter	30	−3.4	−10.6	−4.1	−10.2	−10.4	−6.1	−4.4
	40	−6.6	−10.6	−4.1	−10.1	−10.4	−11.5	−8.4
	50	−10.4	−10.5	−4.0	−10.0	−10.4	−13.2	−12.3
	5	0.6	0.5	0.3	0.5	−0.8	0.6	0.8
	10	0.5	−0.6	−1.8	−1.1	−3.2	0.3	0.7
	20	−1.5	−3.0	−4.0	−3.4	−9.1	−2.4	−1.8
Spring	30	−2.9	−5.6	−7.3	−7.3	−14.1	−3.9	−3.1
	40	−4.3	−10.9	−7.3	−13.2	−14.1	−6.9	−5.2
	50	−6.5	−13.7	−7.3	−13.2	−14.0	−10.7	−7.1

Table 6
Semi-arid city results for PV electric energy production on PV-PCM system for seven PCM with thickness between 5 and 50 mm.

Seasons	mm	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
	5	−0.005%	0.2%	0.0%	0.3%	0.9%	0.1%	−0.08%
	10	0.4%	0.8%	0.4%	1.0%	2.0%	0.5%	0.4%
	20	1.4%	2.1%	1.3%	2.4%	4.3%	1.8%	1.3%
Summer	30	2.1%	3.5%	2.2%	3.8%	6.1%	2.9%	1.8%
	40	2.5%	4.3%	2.8%	4.5%	6.5%	3.4%	2.2%
	50	2.8%	4.6%	3.1%	4.8%	6.5%	3.8%	2.5%
	5	0.0%	0.3%	0.1%	0.3%	1.0%	0.1%	−0.04%
	10	0.5%	0.9%	0.5%	1.1%	2.3%	0.6%	0.5%
	20	1.6%	2.4%	1.6%	2.8%	4.9%	2.1%	1.5%
Autumn	30	2.4%	3.9%	2.6%	4.4%	6.8%	3.3%	2.0%
	40	2.8%	4.9%	3.3%	5.1%	7.2%	3.9%	2.5%
	50	3.2%	5.2%	3.7%	5.4%	7.2%	4.3%	2.8%
	5	0.3%	0.6%	0.1%	0.7%	1.4%	0.4%	0.3%
	10	1.0%	1.4%	0.3%	1.6%	2.8%	1.2%	1.0%
	20	2.4%	3.0%	0.5%	3.1%	3.2%	2.8%	2.7%
Winter	30	3.9%	3.4%	0.6%	3.1%	3.4%	4.3%	4.3%
	40	5.0%	3.4%	0.6%	3.1%	3.6%	4.9%	4.8%
	50	5.3%	3.4%	0.6%	3.1%	3.7%	4.8%	5.1%
	5	0.1%	0.3%	0.1%	0.4%	1.1%	0.2%	0.1%
	10	0.6%	1.1%	0.5%	1.2%	2.4%	0.8%	0.6%
	20	1.9%	2.5%	1.4%	2.8%	4.7%	2.3%	2.1%
Spring	30	3.1%	3.9%	1.8%	4.2%	5.0%	3.7%	3.4%
	40	4.1%	4.8%	1.8%	4.6%	5.1%	4.7%	3.9%
	50	4.5%	4.9%	1.8%	4.6%	5.2%	5.1%	4.3%

electricity production is calculated as the mean energy production of the four seasons studied. In both cities the optimal PV-PCM configuration was 40 mm of $CaCl_2$–$6H_2O$, with an electricity generation that increased 5.8% for the semi-arid city and 4.5% for the arid city.

Analyzing the optimal PV-PCM systems for each climatic zone with 40 mm of $CaCl_2$–$6H_2O$, it was found that the system in the semi-arid zone had a 30% higher yearly electricity generation gain compared to the arid zone. This is attributed to the semi-arid zone operates at higher panel temperatures in the summer and autumn seasons than the arid zone (the two months with the highest power generation). This is attributed to the fact that the semi-arid region has a lower magnitude of wind speeds and higher air temperature than the arid region.

As a preliminary observation of this analysis, we found that the optimal thickness for PV-PCM must be studied for each location. The use of a low PCM thickness (up to 5 mm) increased the PV temperature in the majority of the cases analyzed. This is because the PV-PCM system increases the thickness of the PV panel and hence its thermal resistance, the heat losses by convection and radiation are not large enough to cool down the PV panel. On the other hand, if the thickness is oversized the melted PCM could not completely solidify during the night, and hence that the PV-PCM cooling effect is not increased. Additionally, is found that the use of a PCM with a too high or low melting temperature also could deteriorate the PV efficiency. In the case that the melting temperature is too high, the PV panel starts to melt the PCM too late on the day, and it does not melt it completely.

Table 7
Arid city results for PV electric energy production on PV-PCM system for seven PCM with thickness between 5 and 50 mm.

Seasons	mm	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
	5	0.02%	0.2%	0.02%	0.3%	0.9%	0.1%	−0.05%
	10	0.4%	0.8%	0.3%	1.0%	1.9%	0.6%	0.4%
	20	1.5%	1.9%	0.9%	2.2%	3.7%	1.8%	1.6%
Summer	30	2.5%	3.0%	1.3%	3.2%	4.5%	2.9%	2.8%
	40	3.3%	3.8%	1.3%	3.9%	4.6%	3.8%	3.3%
	50	3.8%	4.2%	1.3%	3.9%	4.7%	4.2%	3.6%
	5	0.003%	0.2%	0.02%	0.3%	0.9%	0.1%	−0.1%
	10	0.4%	0.8%	0.3%	1.0%	2.0%	0.6%	0.4%
	20	1.5%	2.0%	0.9%	2.2%	3.8%	1.8%	1.7%
Autumn	30	2.5%	3.0%	1.4%	3.3%	4.6%	3.0%	2.8%
	40	3.4%	3.9%	1.4%	4.0%	4.7%	3.9%	3.4%
	50	3.9%	4.3%	1.4%	4.0%	4.8%	4.3%	3.7%
	5	0.1%	0.3%	−0.01%	0.4%	1.1%	0.2%	0.1%
	10	0.6%	1.0%	0.2%	1.1%	2.2%	0.8%	0.7%
	20	1.8%	2.2%	0.3%	2.4%	3.1%	2.1%	2.0%
Winter	30	2.9%	3.0%	0.4%	2.9%	3.3%	3.2%	3.3%
	40	3.9%	3.1%	0.4%	2.9%	3.4%	4.1%	4.2%
	50	4.6%	3.1%	0.5%	2.9%	3.6%	4.3%	4.7%
	5	0.0%	0.3%	0.03%	0.3%	1.0%	0.1%	−0.03%
	10	0.5%	0.9%	0.4%	1.1%	2.1%	0.7%	0.5%
	20	1.6%	2.2%	1.1%	2.4%	4.1%	2.0%	1.8%
Spring	30	2.8%	3.3%	1.6%	3.5%	4.8%	3.2%	3.1%
	40	3.8%	4.2%	1.6%	4.3%	4.9%	4.3%	4.1%
	50	4.6%	4.5%	1.6%	4.2%	5.0%	4.9%	4.6%

Table 8
Semi-arid and Arid cities results for yearly PV electric energy production on PV-PCM system for seven PCMs with thickness between 5 and 50 mm.

Region	mm	RT-21	RT-25	RT-35	nOct	$CaCl_2$–$6H_2O$	CA	CL
	5	0.3%	0.1%	0.1%	0.4%	1.0%	0.2%	0.04%
	10	1.0%	0.5%	0.6%	1.2%	2.3%	0.7%	0.6%
	20	2.5%	1.2%	1.7%	2.7%	4.4%	2.2%	1.8%
Semi-arid	30	3.7%	1.9%	2.8%	3.9%	5.6%	3.5%	2.7%
	40	4.4%	2.3%	3.4%	4.5%	5.8%	4.1%	3.1%
	50	4.6%	2.5%	3.8%	4.6%	5.8%	4.4%	3.5%
	5	0.3%	0.01%	0.05%	0.3%	1.0%	0.1%	−0.02%
	10	0.9%	0.3%	0.5%	1.0%	2.1%	0.7%	0.5%
	20	2.1%	0.8%	1.6%	2.3%	3.7%	1.9%	1.8%
Arid	30	3.1%	1.2%	2.7%	3.2%	4.4%	3.1%	3.0%
	40	3.8%	1.2%	3.6%	3.8%	4.5%	4.0%	3.8%
	50	4.1%	1.2%	4.2%	3.8%	4.5%	4.4%	4.1%

On the other hand, if the melting temperature is too low, the PV panel melts the PCM too early in the day.

8. Conclusions

This research studied the use of seven PCMs as a potential passive cooling system for a PV panel under day–night time-varying atmospheric condition cycles (one cycle per season) by numerical simulations that include transient heat losses by convection and radiation. The study was developed for two Chilean cities with high solar irradiation, Vicuña and Calama, which corresponds to a semi-arid and an arid climatic zones, respectively. A simplified one-dimensional PV model with an Enhanced Conduction Model (ECM) – to accomplish a parametric analysis of the heat convection on the melted PCM – were implemented with the Finite Volume Method (FVM) in Julia language. The conclusions founded were:

- The simplified one-dimensional PV model with the ECM proved to describe accurately the PV-PCM system under time-varying atmospheric conditions. The predicted result for the maximum deviation of the PV temperature against the numerical value reported in the literature was 1.23 °C.
- The use of a PV-PCM system reduces the maximum temperature on the PV module and increases the electricity generation. However, the selection of the PCM and its thickness for the PV-PCM system must consider the variability of the atmospheric conditions during the day–night cycle in the four seasons as key design parameters. In this study the design of a suitable PV-PCM system allowed that the maximum temperature decreases up to 17.5°C with an increment of the yearly electricity production of 5.8%.
- The best PV-PCM configuration found for the semi-arid and arid regions consists of 40 mm of $CaCl_2$–$6H_2O$, with an increment in the yearly electricity production of 5.8% and 4.5%, respectively. However, for the semi-arid city was found that the RT-21 had a better thermal performance for the month of June (winter). Although due to the low electricity production at the winter season, the selection of $CaCl_2$–$6H_2O$ had a better performance in the yearly electricity generation.
- The use of a PV-PCM system with 40 mm of $CaCl_2$–$6H_2O$ has a 30% higher yearly electricity generation gain in the semi-arid region than in the arid region. The difference in performance between the two regions is attributed to the fact that the base operating temperature of the panel is higher in the months of highest energy production (summer and autumn) due to high air temperatures and low wind speeds. This highlights the importance of convective heat losses as the main cooling mechanism of the panel.
- A PV-PCM system based on the use of 40 mm of $CaCl_2$–$6H_2O$ should have a better cost effectiveness in a semi-arid region, because the thermal performance gains was found to be higher, then a reduction in the panel area could be expected.

- The methodology proposed for the selection of the optimal PCM and thickness for the PV-PCM system can be extended to other solar devices. This is because the main parameter for the material selection is the time variation of the atmospheric conditions, at the chosen location. Some solar technologies that can benefit from the use of PCMs are: solar collectors for water heating, solar energy accumulators for food dryers, and Trombe walls. The use of PCM in these devices would contribute to reduce the thermal amplitude during the day–night cycle, the displacement of the maximum temperature, and to extend the period of operation of the device. Present results for the PV-PCM system can be used as a preliminary guide for these solar devices under similar atmospheric conditions.

CRediT authorship contribution statement

Felipe A. Díaz: Conceptualization, Methodology, Software, Validation, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Nelson O. Moraga:** Methodology, Validation, Investigation, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Roberto C. Cabrales:** Methodology, Validation, Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Founding

The National Agency of Research and Development - Chile (ANID) under grant to FONDECYT 1200572 project and by the Doctoral Scholarship Program, Chile, grant 21211434, and the supercomputing infrastructure of the NLHPC (ECM-02), partially supported this work.

References

[1] IEA. Key world energy statistics 2021. Key World Energy Stat 2021. http://dx.doi.org/10.1787/2ef8cebc-en.
[2] IEA. Key world energy statistics 2020. Key World Energy Stat. 2020. http://dx.doi.org/10.1787/295f00f5-en.
[3] Dubey S, Sarvaiya JN, Seshadri B. Temperature dependent photovoltaic (PV) efficiency and its effect on PV production in the world – A review. Energy Procedia 2013;33:311–21. http://dx.doi.org/10.1016/j.egypro.2013.05.072.
[4] Tao M, Zhenpeng L, Jiaxin Z. Photovoltaic panel integrated with phase change materials (PV-PCM): technology overview and materials selection. Renew Sustain Energy Rev 2019;116:109406. http://dx.doi.org/10.1016/j.rser.2019.109406.
[5] Khanna S, Reddy K, Mallick TK. Optimization of solar photovoltaic system integrated with phase change material. Sol Energy 2018;163:591–9. http://dx.doi.org/10.1016/j.solener.2018.01.002.
[6] Khanna S, Reddy K, Mallick TK. Optimization of finned solar photovoltaic phase change material (finned PV PCM) system. Int J Therm Sci 2018;130:313–22. http://dx.doi.org/10.1016/j.ijthermalsci.2018.04.033.
[7] Kamkari B, Shokouhmand H, Bruno F. Experimental investigation of the effect of inclination angle on convection-driven melting of phase change material in a rectangular enclosure. Int J Heat Mass Transfer 2014;72:186–200. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2014.01.014.
[8] Nasef H, Hassan H, Mori S, Nada S. Experimental investigation on the performance of parallel and staggered arrays of PCM energy storage system for PV thermal regulation. Energy Convers Manage 2022;254:115143. http://dx.doi.org/10.1016/j.enconman.2021.115143.
[9] Nižetić S, Arıcı M, Bilgin F, Grubišić-Čabo F. Investigation of pork fat as potential novel phase change material for passive cooling applications in photovoltaics. J Cleaner Prod 2018;170:1006–16. http://dx.doi.org/10.1016/j.jclepro.2017.09.164.
[10] Akshayveer Kumar A, Singh AP, Kotha RS, Singh O. Thermal energy storage design of a new bifacial PV/PCM system for enhanced thermo-electric performance. Energy Convers Manage 2021;250:114912. http://dx.doi.org/10.1016/j.enconman.2021.114912.
[11] Emam M, Ahmed M. Cooling concentrator photovoltaic systems using various configurations of phase-change material heat sinks. Energy Convers Manage 2018;158:298–314. http://dx.doi.org/10.1016/j.enconman.2017.12.077.
[12] Stalin PMJ, Prasad KS, Kumar KP, Hemadri G, Rajesh M, Kumar KP. Performance improvement of solar PV through the thermal management using a nano-PCM. Mater Today Proc 2022;50:1553–8. http://dx.doi.org/10.1016/j.matpr.2021.09.111.
[13] Nada S, El-Nagar D, Hussein H. Improving the thermal regulation and efficiency enhancement of PCM-integrated PV modules using nano particles. Energy Convers Manage 2018;166:735–43. http://dx.doi.org/10.1016/j.enconman.2018.04.035.
[14] Abdelrahman H, Wahba M, Refaey H, Moawad M, Berbish N. Performance enhancement of photovoltaic cells by changing configuration and using PCM (RT35hc) with nanoparticles al2o3. Sol Energy 2019;177:665–71. http://dx.doi.org/10.1016/j.solener.2018.11.022.
[15] Al-Waeli AH, Sopian K, Chaichan MT, Kazem HA, Ibrahim A, Mat S, Ruslan MH. Evaluation of the nanofluid and nano-PCM based photovoltaic thermal (PVT) system: An experimental study. Energy Convers Manage 2017;151:693–708. http://dx.doi.org/10.1016/j.enconman.2017.09.032.
[16] Brano VL, Ciulla G, Piacentino A, Cardona F. Finite difference thermal model of a latent heat storage system coupled with a photovoltaic device: Description and experimental validation. Renew Energy 2014;68:181–93. http://dx.doi.org/10.1016/j.renene.2014.01.043.
[17] Arıcı M, Bilgin F, Nižetić S, Papadopoulos AM. Phase change material based cooling of photovoltaic panel: A simplified numerical model for the optimization of the phase change material layer and general economic evaluation. J Cleaner Prod 2018;189:738–45. http://dx.doi.org/10.1016/j.jclepro.2018.04.057.
[18] Velmurugan K, Kumarasamy S, Wongwuttanasatian T, Seithtanabutara V. Review of PCM types and suggestions for an applicable cascaded PCM for passive PV module cooling under tropical climate conditions. J Cleaner Prod 2021;293:126065. http://dx.doi.org/10.1016/j.jclepro.2021.126065.
[19] Moraga NO, Xamán JP, Araya RH. Cooling Li-ion batteries of racing solar car by using multiple phase change materials. Appl Therm Eng 2016;108:1041–54. http://dx.doi.org/10.1016/j.applthermaleng.2016.07.183.
[20] Elsheniti MB, Hemedah MA, Sorour M, El-Maghlany WM. Novel enhanced conduction model for predicting performance of a PV panel cooled by PCM. Energy Convers Manage 2020;205:112456. http://dx.doi.org/10.1016/j.enconman.2019.112456.
[21] Smith CJ, Forster PM, Crook R. Global analysis of photovoltaic energy output enhanced by phase change material cooling. Appl Energy 2014;126:21–8. http://dx.doi.org/10.1016/j.apenergy.2014.03.083.
[22] Rubitherm G. Rubitherm data sheet. 2000, Hamburg: Rubitherm GmbH.
[23] Biwole PH, Eclache P, Kuznik F. Phase-change materials to improve solar panel's performance. Energy Build 2013;62:59–67. http://dx.doi.org/10.1016/j.enbuild.2013.02.059.
[24] Huang M, Eames P, Norton B, Hewitt N. Natural convection in an internally finned phase change material heat sink for the thermal management of photovoltaics. Sol Energy Mater Sol Cells 2011;95(7):1598–603. http://dx.doi.org/10.1016/j.solmat.2011.01.008.
[25] Qarnia HE. Numerical analysis of a coupled solar collector latent heat storage unit using various phase change materials for heating the water. Energy Convers Manage 2009;50(2):247–54. http://dx.doi.org/10.1016/j.enconman.2008.09.038.
[26] Hasan A, McCormack S, Huang M, Norton B. Characterization of phase change materials for thermal control of photovoltaics using differential scanning calorimetry and temperature history method. Energy Convers Manage 2014;81:322–9. http://dx.doi.org/10.1016/j.enconman.2014.02.042.
[27] Mattei M, Notton G, Cristofari C, Muselli M, Poggi P. Calculation of the polycrystalline PV module temperature using a simple method of energy balance. Renew Energy 2006;31(4):553–67. http://dx.doi.org/10.1016/j.renene.2005.03.010.
[28] Messenger R, Abtahi A. Photovoltaic systems engineering. fourth ed.. CRC Press; 2017, p. 53. http://dx.doi.org/10.1201/9781315151434, pp. 53.
[29] Moraga NO, Marambio MA. Three-dimensional natural heat convection and ternary alloy solidification problems by finite volume geometric multigrid method. Numer Heat Transf A: Applications 2020;77(6):632–48. http://dx.doi.org/10.1080/10407782.2020.1713685.
[30] Hollands KGT, Unny TE, Raithby GD, Konicek L. Free convective heat transfer across inclined air layers. J Heat Transfer 1976;98(2):189–93. http://dx.doi.org/10.1115/1.3450517.
[31] Verkovsky BM, Polevikov VK. Numerical study of problems on high-intensive free convection. Heat Transf Turbulent Buoyant Convection 1977;443–55.
[32] MacGregor RK, Emery AF. Free convection through vertical plane layers—moderate and high prandtl number fluids. J Heat Transfer 1969;91(3):391–401. http://dx.doi.org/10.1115/1.3580194.

[33] Catton I. Natural convection in enclosures. In: International heat transfer conference digital library. Begel House Inc.; 1978.
[34] Zhou J, Yi Q, Wang Y, Ye Z. Temperature distribution of photovoltaic module based on finite element simulation. Sol Energy 2015;111:97–103. http://dx.doi.org/10.1016/j.solener.2014.10.040.
[35] Loveday D, Taki A. Convective heat transfer coefficients at a plane surface on a full-scale building facade. Int J Heat Mass Transfer 1996;39(8):1729–42. http://dx.doi.org/10.1016/0017-9310(95)00268-5.
[36] Swinbank WC. Long-wave radiation from clear skies. Q J R Meteorol Soc 1963;89(381):339–48. http://dx.doi.org/10.1002/qj.49708938105.
[37] Molina A, Falvey M, Rondanelli R. A solar radiation database for Chile. Sci Rep 2017;7(1):14823. http://dx.doi.org/10.1038/s41598-017-13761-x.
[38] Bezanson J, Edelman A, Karpinski S, Shah VB. Julia: A fresh approach to numerical computing. SIAM Rev 2017;59(1):65–98. http://dx.doi.org/10.1137/141000671.
[39] Patankar SV. Numerical heat transfer and fluid flow vol. 58. Boca Raton: CRC Press; 1980, p. 39–41.
[40] Davis TA. Algorithm 832: UMFPACK V4.3—an unsymmetric-pattern multifrontal method. ACM Trans Math Softw 2004;30(2):196–9. http://dx.doi.org/10.1145/992200.992206.
[41] Solomon A. An easily computable solution to a two-phase stefan problem. Sol Energy 1979;23(6):525–8. http://dx.doi.org/10.1016/0038-092x(79)90077-x.